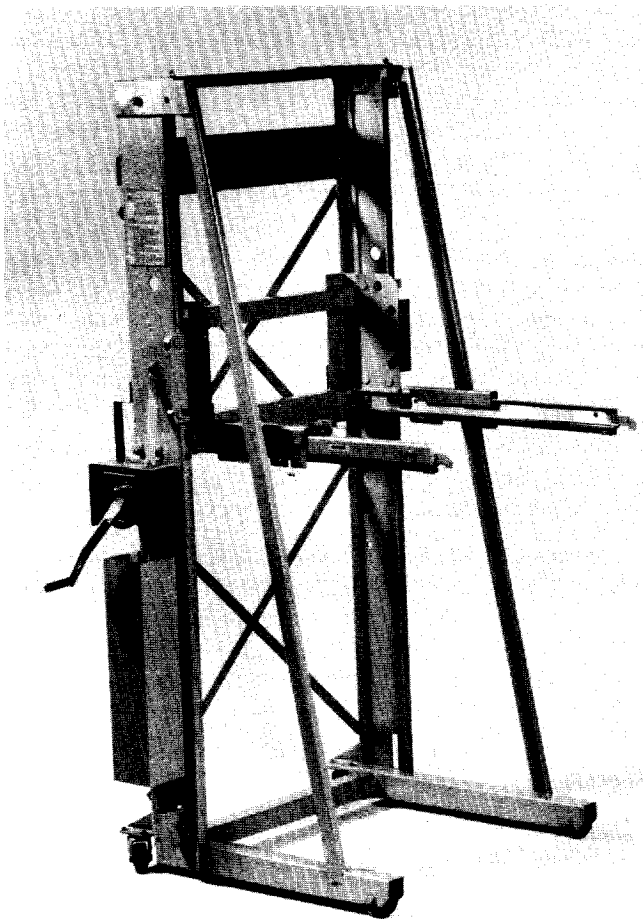
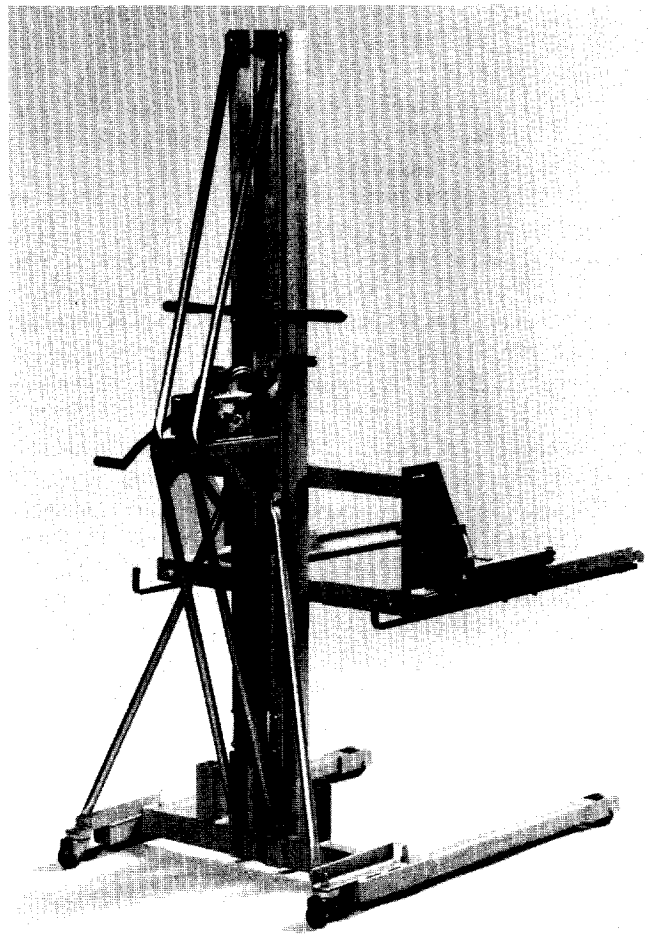
INSTRUCTIONS

GEK 90214 A

GE LIFT TRUCK





GE LIFT TRUCK

EACH USER HAS THE RESPONSIBILITY TO INSTRUCT ALL PERSONNEL ASSOCIATED WITH THIS EQUIPMENT ON ALL SAFETY PRECAUTIONS WHICH MUST BE OBSERVED.

GENERAL

GE offers two styles of lift trucks for handling Power/Vac circuit breakers, ground and test devices and transformer and fuse rollouts. The first is a double-masted truck that is available with four swivel casters (0144D2912G001) or with two swivel casters in the rear and two straight wheels in the front (0144D2912G002). This truck is compatible with indoor switchgear. The second style is a single-masted truck (0144D2911G001) that is available only with two swivel casters and two straight wheels. This truck is compatible with indoor and outdoor switchgear.

The lift trucks are provided with interlocks to retain the device being handled and to lock the lift truck to the switchgear while a device is being inserted or removed. The carriage which lifts a device is raised or lowered by means of a winch and cable. When the winch handle is released the carriage is held in that position by means of a clutch-brake internal to the winch.

Two arms are attached to the carriage for engaging the track rollers on the sides of each device. The following procedures describe the necessary steps to remove a device from, or insert a device into, the switchgear equipment. Be sure to read and understand the safety rules identified in this manual before operating the lift truck.

The lift trucks are functional for both upper and lower compartments provided the equipment is mounted on no more than a three inch pad.

NOTE: The figures used in this instruction book depict the double-masted lift truck. The operating instructions are the same for both trucks.

TO REMOVE A DEVICE FROM THE SWITCHGEAR EQUIPMENT:

1. Prepare the truck for docking with the switchgear equipment by making sure that both track roller interlocks (1, Fig. 1) are in the "Transport" position (Fig. 2). If they are not, depress the track roller interlocks and slide the docking interlocks (2, Fig. 1) all the way back toward the lift truck masts.

2. Adjust the height of the carriage until the docking hooks at the ends of the arms are slightly higher than the rail surface in the equipment. Position the hooks over the slots in the rails and lower the hooks into the slots. Place both track roller interlocks (1, Fig. 1) in the "Load/Unload" position (Fig. 3) by depressing them and sliding the docking interlocks (2, Fig. 1) forward, away from the lift truck masts. The docking interlocks will extend under the equipment rails and lock the truck arms to the rails. The track roller interlocks are now in the raised position so they are no longer blocking the track rollers. Use the winch to adjust the lift truck's arms if they are not approximately level.

3. Release the device from the equipment and roll it fully onto the lift truck rails until it comes in contact with the stops (3, Fig. 7) on the rails. The catches that retain a device in the equipment compartment are different for the different devices so see the instructions for the device being handled for specific information. Verify that the device rolled freely and that its rollers have properly engaged the rails. Place both track roller interlocks (1, Fig. 1) in the "Transport" position (Fig. 2) by depressing them and sliding the docking interlocks (2, Fig. 1) all the way back toward the lift truck masts. Raise the carriage slightly to release the docking hooks from the slots in the equipment rail. Pull the lift truck back to just clear the equipment and lower the carriage to, or below, the maximum transport position (Fig. 4) before transporting the device to the desired location.

4. To unload the device onto the floor, lower the carriage until the device is resting on the floor and the load of the track rollers is no longer being supported by the lift truck arms. Place both track roller interlocks (1, Fig. 1) in the "Load/Unload" position (Fig. 3) by depressing them and sliding the docking interlocks (2, Fig. 1) forward, away from the lift truck masts. The track roller interlocks are now in their raised position. The lift truck can now be pulled away from the device while holding the backup track roller interlock handle (4, Fig. 1) in the released position, away from the lift truck masts.

TO INSTALL A DEVICE INTO THE SWITCHGEAR EQUIPMENT:

1. To load a device that is on the floor onto the lift truck place both track roller interlocks (1, Fig. 1) in the "Load/Unload" position (Fig. 3) by depressing them and sliding the docking interlocks (2, Fig. 1) forward, away from the lift truck masts. Adjust the height of the arms so that they can engage the track rollers on the sides of the device. Slight read-

justments may be necessary as the arms approach the rollers. It is also common to have to make minor adjustments to allow the second set of track rollers to be engaged. Move the lift truck forward, or pull the device back, so that the device is all the way towards the lift truck masts and the track rollers are in contact with the stops on the rails. Verify that the device rolled freely and that all rollers have properly engaged the rails.

2. To lock the device onto the lift truck, place both track roller interlocks (1, Fig. 1) into the "Transport" position (Fig. 2) by depressing them down and sliding the docking interlocks (2, Fig. 1) all the way back toward the lift truck masts.

3. Raise the carriage so the device is off the floor but the carriage is at, or below, the maximum transport height before moving the device into position directly in front of the equipment compartment into which the device is to be installed. The lift truck should be as close to the equipment as possible, allowing just enough room for the device and carriage to clear the equipment as it is being raised. Adjust the height of the carriage until the docking hooks at the ends of the arms are slightly higher than the rail surface in the equipment. Position the hooks over the slots in the rails and lower the hooks into the slots. Place both track roller interlocks (1, Fig. 1) in the "Load/Unload" position (Fig. 3) by depressing them and sliding the docking interlocks (2, Fig. 1) forward, away from the lift truck masts. The docking interlocks will extend under the equipment rails and lock the truck arms to the rails. The track roller interlocks are now in the raised position so they are no longer blocking the track rollers. Use the winch to adjust the lift truck's arms if they are not approximately level.

4. Roll the device into the equipment while holding the backup track roller interlock handle (4, Fig. 1) in the released position, away from the lift truck masts. Make sure that the device has engaged the latches in the equipment (see the instructions for the equipment and the device being used).

5. To undock the lift truck from the equipment place both track roller interlocks (1, Fig. 1) in the "Transport" position (Fig. 2) by depressing them and sliding the docking interlocks (2, Fig. 1) all the way back toward the lift truck masts. Raise the carriage slightly to release the docking hooks from the slots in the equipment rail. Pull the lift truck back to clear the equipment and lower the carriage to, or below, the maximum transport position (Fig. 4).

WARNING

As the load is lifted there should be a clicking noise from the winch pawl. If this noise stops while a device is being lifted, maintain a grip on the handle and lower the load to the floor. Investigate why the pawl is not engaging and make repairs before making any further attempt to lift the load.

CAUTION

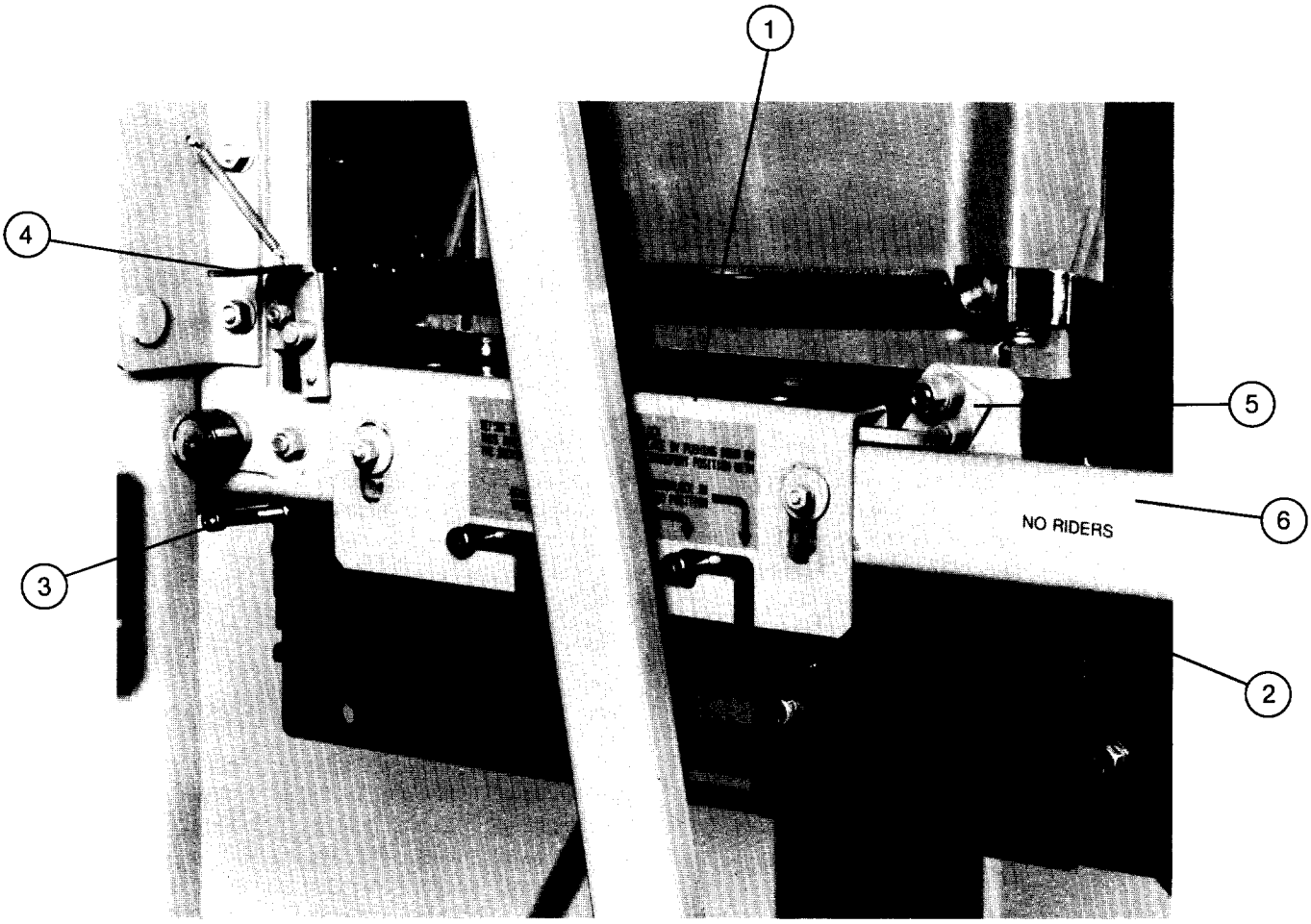
Although a backup to the track roller interlock is functional for all but long wheel base devices such as electrical ground & test devices and fuse rollouts, safe use of this truck demands locking of both track roller interlocks in the transport position before winch operation.

SAFETY RULES

1. Maximum load is 850 pounds.
2. Inspect lift truck daily before use. Items inspected must include cable, cable fittings, pulleys, pulley hardware and winch.
3. Replace damaged or frayed cable.
4. Stay out from under elevated load.
5. Lower lift truck carriage to or below maximum transport height before transporting.
6. Use on smooth level surface only.
7. When lowering carriage, stop winch when cable goes slack to prevent tangling.
8. The use of this lift truck for anything other than handling GE Power/Vac devices can cause damage to the lift truck and present a safety hazard.

The GE lift truck is included under the Occupational Safety and Health Administration (OSHA) standard 1910.179 and should be inspected and tested per that standard.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to GE.



1. Track Roller Interlock (Right Side)
2. Docking Interlock
3. Docking Interlock Handle
4. Backup Track Roller Interlock Handle
5. Backup Track Roller Interlock
6. Carriage Arm

Figure 1. Right Side Carriage Arm Assembly

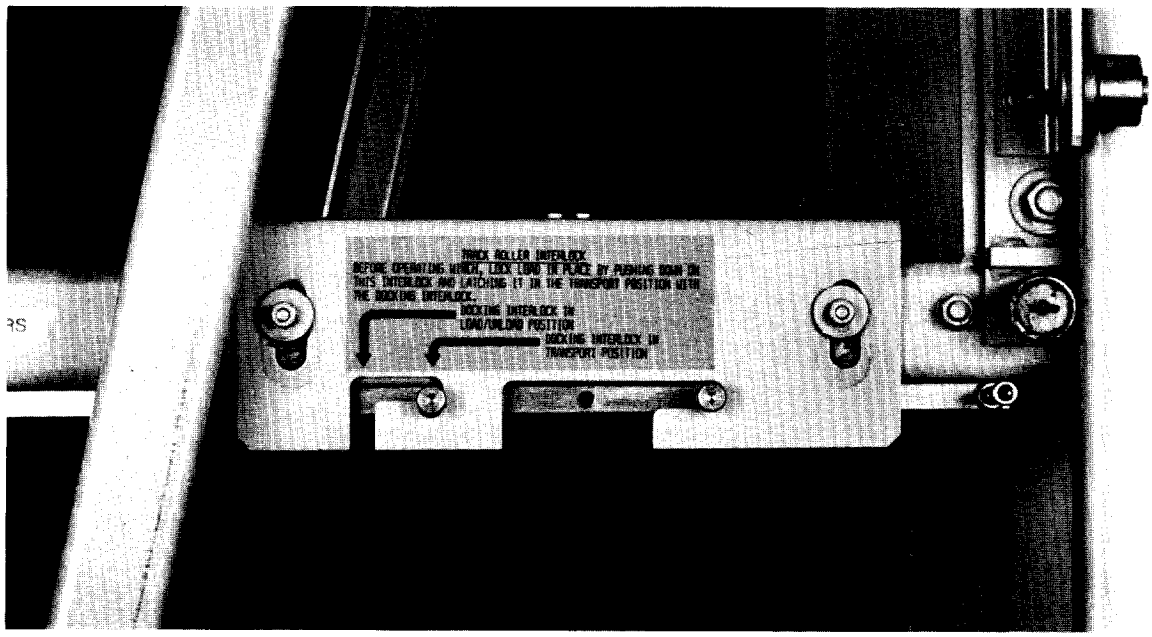


Figure 2. Track Roller and Docking Interlocks in Transport Position

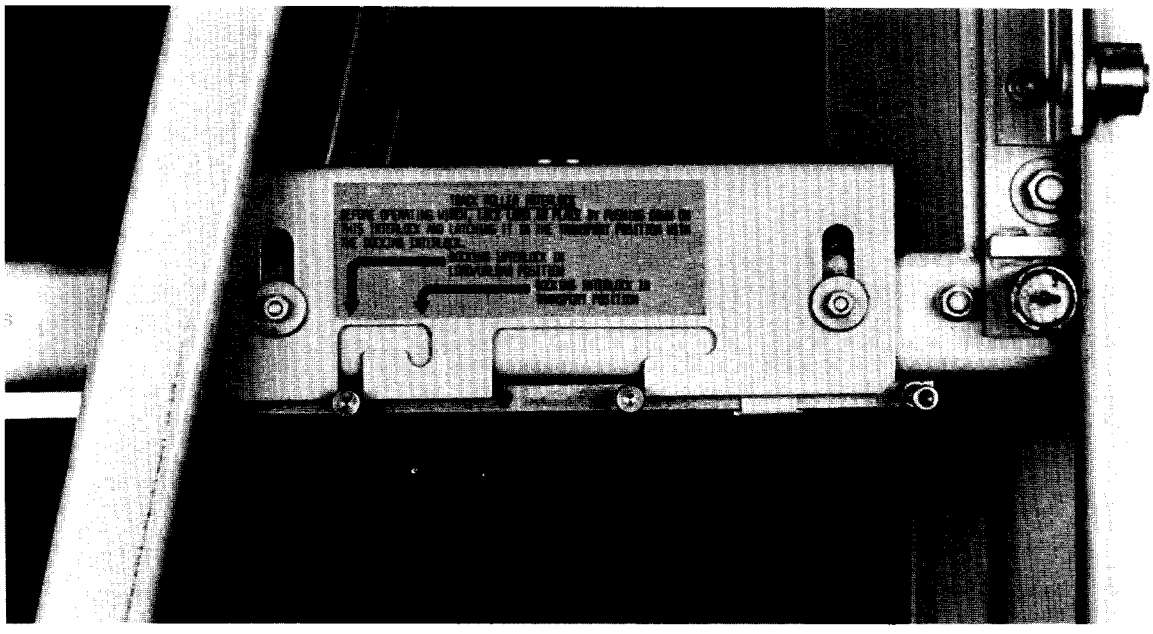


Figure 3. Track Roller and Docking Interlocks in Load/Unload Position

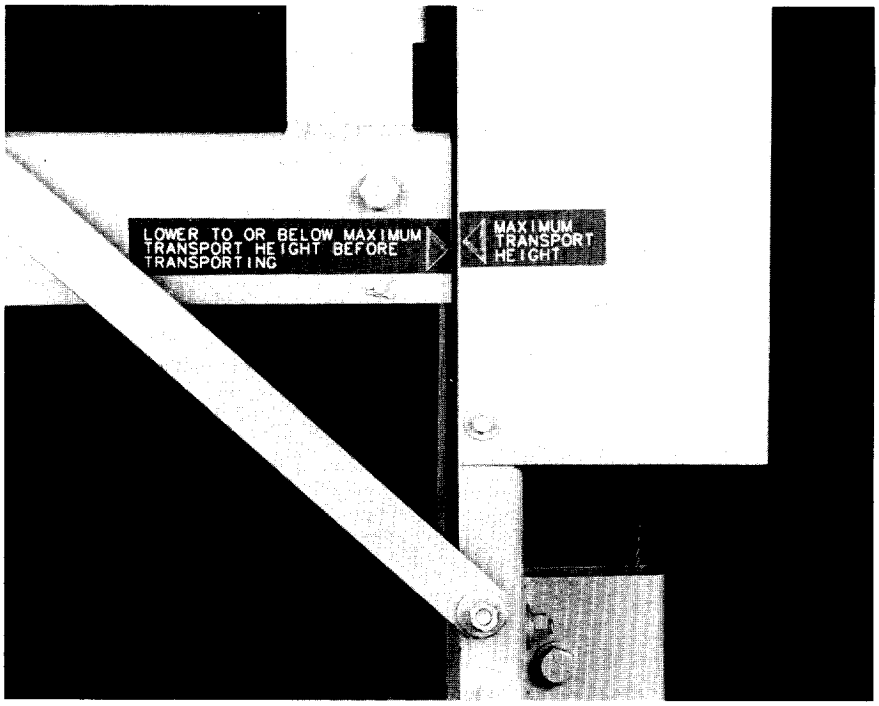
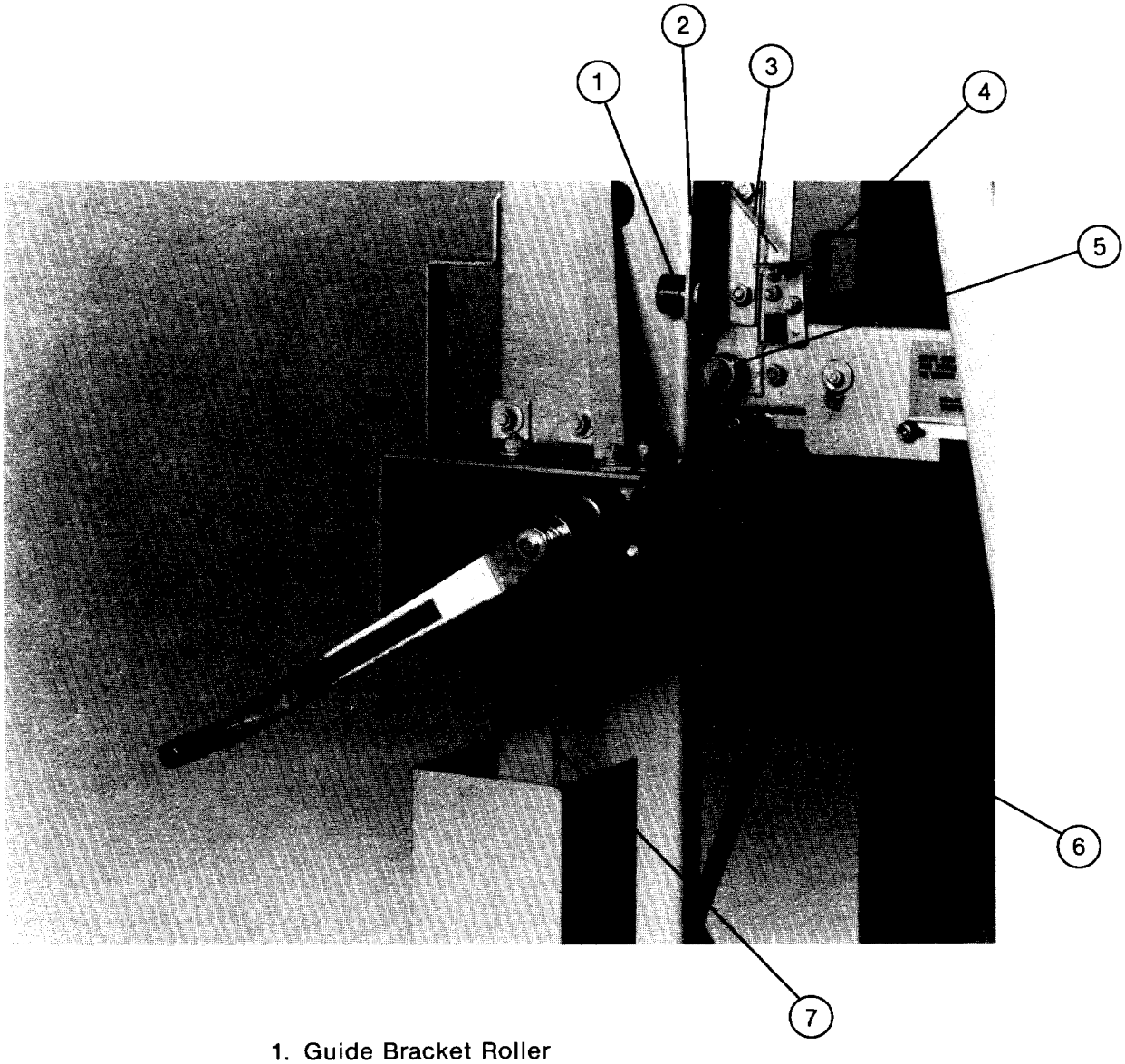


Figure 4. Maximum Transport Height.

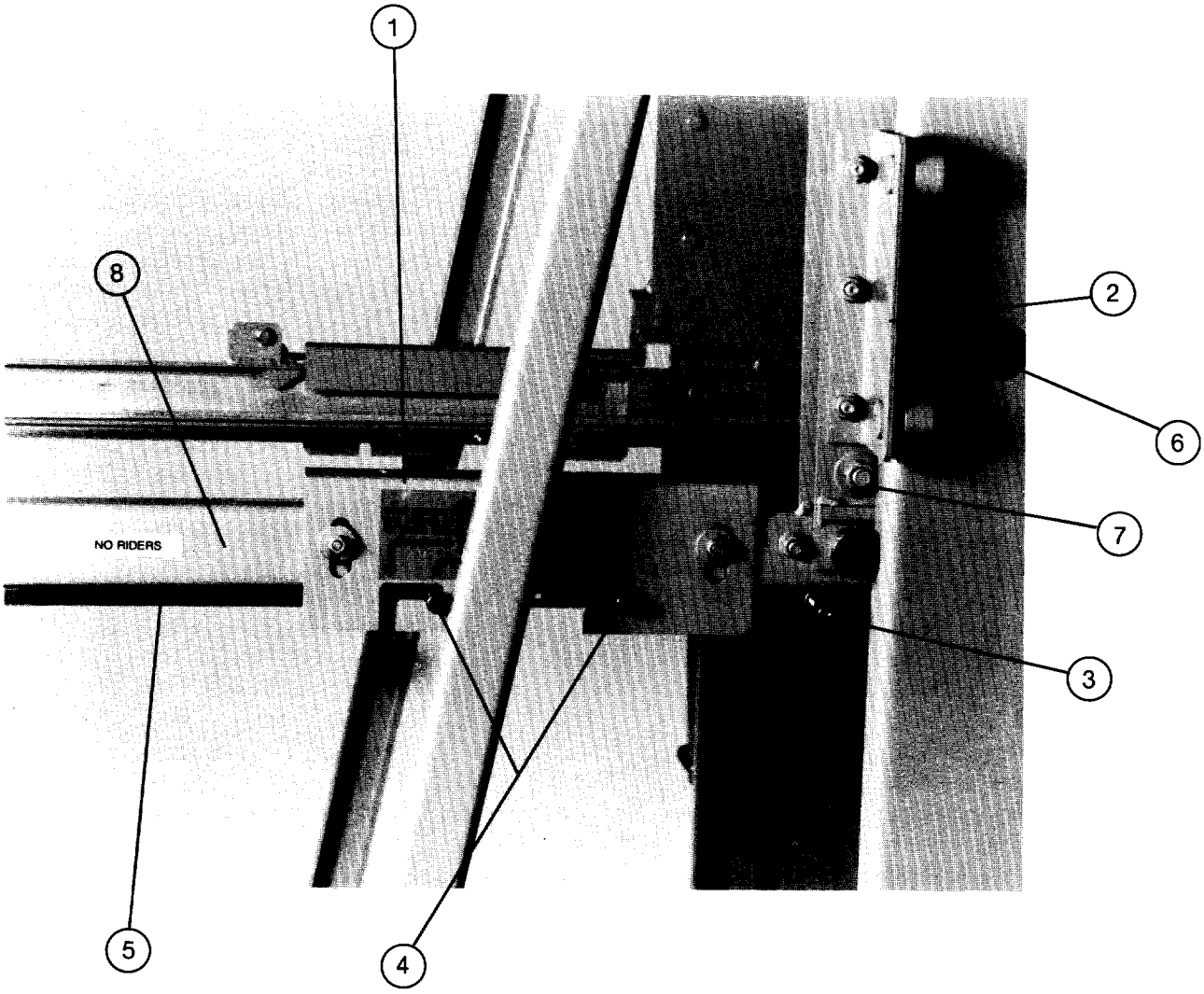
Table 1. Renewal Parts

Fig	Item	Description	Double Masted Lift Truck (0144D2912G001&2) Part Number	Single Masted Lift Truck (0144D2911G001) Part Number
1	1	Track Roller Interlock (Right Side)	0282A4397G004	0282A4397G004
5	1	Guide Roller	0282A3747P001	
	2	Guide Bracket (Right Side)	0282A4161G001	
	3	Backup Track Roller Interlock Spring	0282A3848P001	0282A3848P001
	4	Backup Track Roller Interlock	0209B4788G001	0209B4788G001
	5	Carriage Roller	0282A4569P001	
	6	Winch	0282A3764G002	
	7	Cable	0282A3575P001	
6	1	Track Roller Interlock (Left Side)	0282A4397G003	0282A4397G003
	2	Guide Bracket (Left Side)	0282A4161G002	
	3	Docking Interlock Handle	N170P23040B6	
	4	Shoulder Bolts	N94P38012B6	N94P38012B6
	5	Docking Interlock	0282A4399P001	0209B5160P001
	8	Left Arm	0209B4780G001	0144D2682G001
7	1	Carriage/Right Arm	0177C3744G001	0144D2682G002
	2	Pulley	0282A4469P001	
		Bearing	0282A4556P001	
		Inner Race	0282A2000P055	
	3	Stop Block	0282A4349P001	0282A4349P001
	4	Track Extrusions (Right Side)	0209B4774P001	0209B4774P001
		Track Extrusions (Left Side)	0209B4774P002	0209B4774P002
-	-	Castor	0282A4555P001	
-	-	Straight Wheel (0144D2912G002 only)	0282A4398P001	



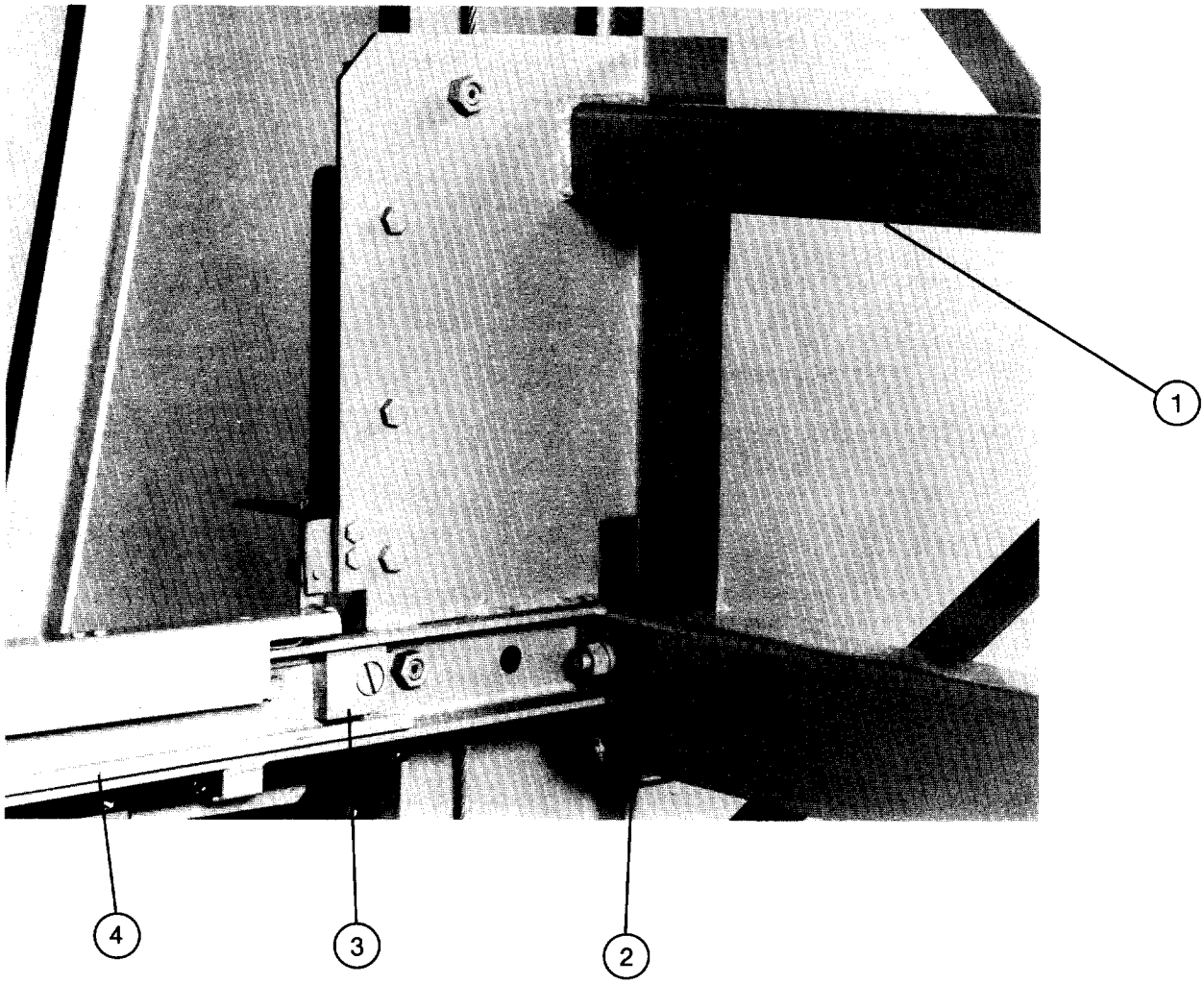
1. Guide Bracket Roller
2. Right Side Guide Bracket
3. Backup Track Roller Interlock Return Spring
4. Backup Track Roller Interlock
5. Carriage Roller
6. Winch
7. Cable

Figure 5. Right Side



1. Track Roller Interlock (Left Side)
2. Left Side Guide Bracket
3. Docking Interlock Handle
4. Shoulder Bolts
5. Docking Interlock
6. Arm Width Adjustment Access Hole
7. Arm Width Adjustment Nut (1 of 3)
8. Left Arm

Figure 6. Left Side



1. Carriage
2. Pulley
3. Stop Block
4. Track Extrusion

Figure 7. Carriage Inside

CARRIAGE ARM ADJUSTMENTS

The arms are adjusted at the factory prior to shipping. Inspect carriage arm spacing if the device being lifted does not fit on lift truck. The distance between the two aluminum track extrusions (4, Fig. 7) should be 29.80 inches, plus 0.00, minus 0.03, and they should be vertically parallel within 0.125 inches.

On the double-masted lift truck the spacing between the arms may be adjusted with three adjusting nuts (7, Fig. 6), two of which must be reached through the access hole in the left mast (6, Fig. 6). The left arm may be adjusted vertically parallel to the right arm by loosening the jam nut and turning the adjusting bolt located underneath the left arm.

The arms are not adjustable vertically on the single-masted lift truck, but they may be slid closer together or farther apart. Enlarge the hole for the carriage arm spring pin if more adjusting distance is required.

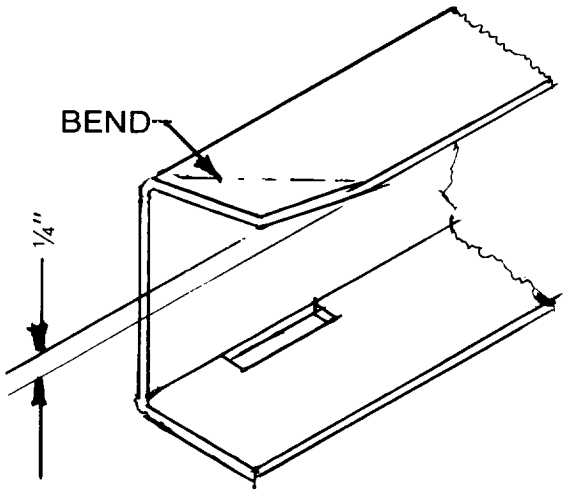
HANDLING OF ROLLOUTS:

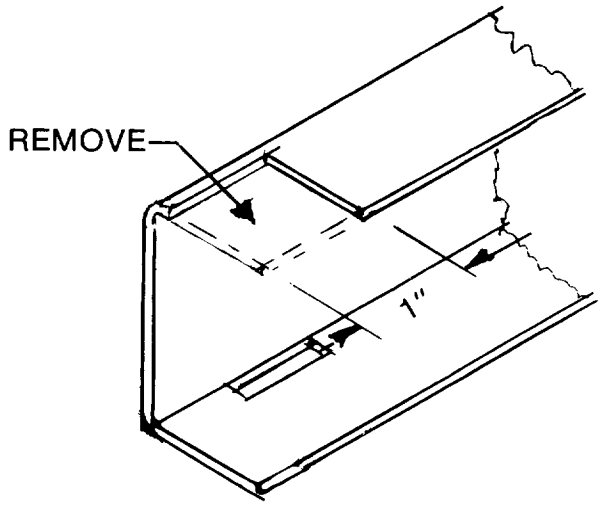
It is possible that on some early fuse and transformer rollouts the rear track rollers may extend about 1.5 inches beyond the end of the lifting arms. Although the front two sets of rollers engaged by the lifting arms provide for safe lifting of the rollout, it may not be possible to dock or undock the lift truck because of interference between the rear rollers and the top of the front end of the equipment track. The arms are prevented from being lifted to release the docking hooks. If this occurs the following action may be required to allow removal of a rollout.

Temporarily remove the stops on the lift truck arms so that the rollout can enter the lifting arms farther. The stops (3, Fig. 7) are rectangular plates on the inside of the "C" shaped lifting arms (6, Fig. 1), held by a single bolt. In many cases this will allow for rollout removal. When reassembling, be sure to place the long end of the part away from the mast. If rollout removal is not possible after this alteration, either of two options will allow removal of the rollout:

1. A 10 or 12 inch adjustable type wrench may be used to slightly bend the top front corner of the equipment rail up approximately 0.25 inch (see illustration below).
2. Remove a piece from the top front of the equipment rail so that the arm can be lifted to release the hook without interfering with the top of the rail (see illustration below). Care must be taken to prevent chips from falling into the equipment.

On new rollouts the rear track rollers have been moved forward 1.5 inches so they are fully engaged by the lift truck arms and there is no interference with the equipment rail.





GE Switchgear Operations
Burlington, IA 52601



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